

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 50

Code of Conduct

I _Narmatha J (192521341)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _Narmatha J_

Assessment Tasks

Class Test

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.
2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.
3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.
4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.
5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.
6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.
7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.
8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.

9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.
10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Class Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

CLASS TEST

6. IMPACT ON DATA QUALITY ISSUES AND DATA GOVERNANCE STRATEGIES :

Data quality is very important in a large-scale analytics platform because poor-quality data can produce incorrect analytical results and wrong business decisions.

MAIN DATA QUALITY ISSUES :

*. Inconsistency :

Same data may be stored in different formats.
For eg., Male, M, and 1.

*. Duplication :

The same record may be stored multiple times, causing incorrect results.

*. Missing values : Some important information may not be available.

*. Incorrect data :

Data may contain wrong values due to human or system errors.

IMPACT :

- *. Produces incorrect analysis
- *. Reduces accuracy of machine-learning models.
- *. Leads to wrong business decisions.
- *. Increase storage and processing costs
- *. Reduces user trust.

DATA GOVERNANCE STRATEGIES

- * Validate data before storing it
- * Remove duplicate records
- * Handle missing values properly
- * Use standard formats
- * Regularly check data quality
- * Protect data using access control and security
- * Assign responsible people for managing data.

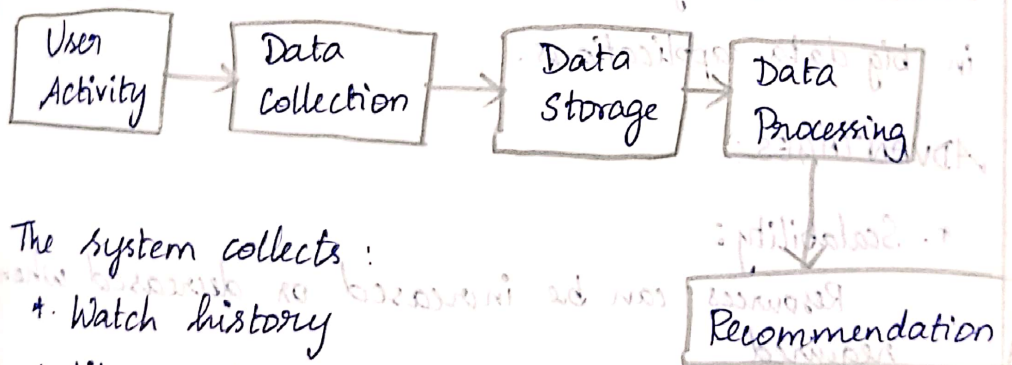
CONCLUSION :

Proper data governance makes data accurate, consistent, complete, secure, and reliable.

7. SCALABLE RECOMMENDATION SYSTEM :

A recommendation system suggests movies, songs, or videos based on user interests.

WORKING :



The system collects :

- * Watch history
- * Likes and dislikes
- * Ratings
- * Search history
- * Watch duration

Large amounts of data can be stored using HDFS or cloud storage. Technologies like Hadoop and Spark process the data using multiple computers.

BENEFITS:

- * Personalized recommendations
- * Better accuracy
- * Faster Processing
- * Increased User engagement
- * More viewing time
- * Better customer satisfaction.

CONCLUSION:

Bigdata and distributed processing help streaming services provide fast, accurate, and personalized recommendations.

8. CLOUD COMPUTING IN BIG DATA:

Cloud Computing provides storage and computing resources through the internet. It is widely used in big data applications.

ADVANTAGES:

* Scalability:

Resources can be increased or decreased when required.

* Cost saving:

Organisation do not need to purchase expensive servers.

* Flexibility:

Different tools and services can be easily used.

LIMITATIONS:

- * Security: Data stored in the cloud may face security threats.
- * Cost: Continuous use can increase the cloud bill.
- * Internet dependency: Internet problems can affect services.
- * Vendor lock-in: Moving from one cloud provider to another can be difficult.
- * Privacy: Sensitive data must be properly protected.

CONCLUSION:

Cloud computing provides scalability, flexibility and storage for big data but security, cost and privacy must be carefully managed.

9. REAL-TIME ANALYTICS FOR TRANSPORTATION:

Real-time analytics helps transportation and logistics companies monitor vehicles and make quick decisions.

FRAMEWORK:



The system collects:

- * Vehicle location
- * Speed
- * Traffic information
- * Fuel consumption
- * Delivery status

Technologies like Apache Kafka can collect real-time data, while Spark Streaming can process it.

Route Optimization:

Traffic information can be analysed to find the best route and avoid congested roads.

Fuel Efficiency:

Vehicle speed, distance and fuel usage can be analyzed to reduce unnecessary fuel consumption.

Decision Making:

Managers can monitor vehicles and quickly respond to delays, accidents or vehicle problems.

BENEFITS:

- * Faster delivery
- * Reduced fuel consumption
- * Less travel time
- * Better vehicle utilization
- * Reduced operating cost
- * Quick decision making

CONCLUSION:

Big data technologies help transportation companies improve routes, fuel efficiency, delivery performance and operational decisions.

10. ETHICAL AND LEGAL CHALLENGES OF BIG DATA:

Big data is useful, but collecting large amounts of personal information creates ethical and legal problems.

MAJOR CHALLENGES:

- * Privacy - Personal information may be misused.
- * Consent - Users may not know what data is being collected
- * Security - Data can be stolen or attacked by hackers.
- * Data Misuse: Data may be used for purposes other than the original purpose.
- * Bias - Biased data can produce unfair results.

RECOMMENDED POLICIES:

- * Obtain user consent
- * Collect only necessary data
- * Clearly explain data usage
- * Encrypt sensitive data
- * Give access only to authorized users

CONCLUSION:

Organizations should use big data responsibly, securely, fairly and legally to protect users and maintain their trust.